

# JORDAN NEWMAN

## Distributed Systems Engineer | Compute Infrastructure, Runtime, Systems Software

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*Selected public repositories are available. The most significant engineering repositories are private; access is available upon request.*

### SUMMARY

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Distributed systems and systems software engineer with more than 20 years of software development and production infrastructure experience. Architected, built, debugged, and operated large-scale compute and networking platforms spanning approximately 3,000-4,000 servers, 10 locations, and more than 65 Gbps of peak traffic. Built a non-blocking FreeBSD HTTP/CDN runtime in C, specialized I/O workers, kernel modules, performance and troubleshooting tools, DDoS defenses, telemetry, deployment automation, and modern Kubernetes, container, and infrastructure-as-code workflows. Production Kubernetes experience is supplemented by hands-on lab work with kubelet, containerd, and runc, plus familiarity with NRI plugin architecture. Strongly familiar with Linux kernel architecture and systems internals, backed by production FreeBSD kernel-module development. Brings hands-on depth from operating-system and networking internals through fleet-scale reliability and operations.

### TECHNICAL SKILLS

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- **Languages:** C, C++, Go, Python, C#, Objective-C, PHP, Perl, JavaScript, Node.js, Shell, Visual Basic
- **Systems and Performance:** Linux kernel architecture and internals, FreeBSD kernel-module development, system calls, processes, memory, networking, filesystems, scheduling, kqueue, sendfile(), GDB, performance debugging, troubleshooting, non-blocking I/O
- **Networking:** TCP/IP, IPv4, sockets, HTTP, DNS, CDN architecture, packet capture, libpcap, IPFW, host networking, authentication, telemetry
- **Cloud and Orchestration:** Kubernetes, Docker, container orchestration, Terraform, Azure DevOps, CI/CD, security scanning, deployment automation; experimental kubelet, containerd, and runc; familiarity with NRI plugin architecture
- **Databases/Web:** MySQL, MultiValue/Universe, OAuth/OAuth2, SMTP OAuth, React, jQuery, PHP, web services

### SELECTED TECHNICAL HIGHLIGHTS

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- Designed and deployed PrimeHTTPD, a custom event-driven compute and HTTP serving runtime across approximately 200 production FreeBSD servers.
- Supported more than 150,000 concurrent connections using kqueue, persistent connections, non-blocking sockets, zero-copy sendfile(), and FreeBSD-specific TCP and accept-filter optimizations.
- Diagnosed and removed runtime bottlenecks by separating latency-sensitive event processing from blocking disk work through dedicated I/O workers and SF\_NODISKIO-aware handling.
- Architected and operated distributed compute, hosting, and CDN infrastructure spanning thousands of servers and multiple geographic locations.
- Built observability, traffic analysis, DNS monitoring, authentication, deployment, security, and DDoS mitigation systems for 24/7 production operations.
- Implemented Kubernetes, Docker, Terraform, Azure DevOps, CI/CD, and security-scanning workflows across modern engineering environments.
- Used kubelet, containerd, and runc in hands-on experiments and studied the fundamentals of developing NRI plugins; this work was exploratory rather than production runtime development.
- Brings strong Linux kernel familiarity and production kernel-development experience on FreeBSD, without claiming Linux kernel-module authorship.

### SELECTED SYSTEMS PROJECTS

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- **PrimeHTTPD:** Architected and implemented a high-performance, non-blocking, kqueue-based HTTP/CDN runtime in C for FreeBSD. Used sendfile(), SF\_NODISKIO, TCP\_NODELAY, TCP\_NOPUSH, O\_NONBLOCK, accept\_filter\_http, persistent connections, and dedicated I/O workers to improve throughput, reliability, and latency under production load.
- **VirtualDir:** Designed and developed a production FreeBSD kernel module that intercepted filesystem-related syscalls and transparently remapped paths from configuration, enabling shared physical layouts without chroot jails; included the vdcli runtime management tool.
- **PrimeDump:** Built a libpcap-based real-time network troubleshooting and traffic-analysis tool decoding Ethernet, IPv4/IPv6, TCP, UDP, and ICMPv6 with ncurses views and IPFW integration.
- **PrimeDNSTop:** Built a DNS traffic monitor parsing UDP/53 and RFC 1035 queries to detect recursion attacks and rank source and domain activity.
- **TAFOS:** Developed an educational x86 operating-system kernel in C and assembly with a custom MBR bootloader, protected mode, IDT, heap allocator, port I/O, VGA output, and GDB support.

### PROFESSIONAL EXPERIENCE

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#### Advantive - Senior Development Specialist

Remote / Tampa Bay, FL | July 2022 - Present

- Implemented Kubernetes, Docker, Terraform, Azure DevOps, CI/CD, and security-scanning workflows across engineering teams.
- Served as a member of the Architecture Team, collaborating with fellow architects and the architecture lead to design technical solutions, define engineering standards, guide platform direction, and develop shared libraries, documentation, and AI-assisted engineering workflows.

#### DDI Systems - Senior Development Specialist

Manalapan, NJ | November 2021 - July 2022

- Modernized legacy VB.NET components in C# and integrated Microsoft Office 365 SMTP OAuth for secure customer-facing workflows.

#### ISPrime LLC - CEO

Weehawken, NJ | April 2018 - January 2020

- Led datacenter modernization, operational restructuring, and improvements to internally developed monitoring, security, and infrastructure platforms.

### DDI Systems - Computer Programmer

Manalapan, NJ | April 2017 - April 2018

- Built ERP integrations and automated order, catalog, image, SDS, SFTP, SellerCloud, Essendant, and MultiValue/Universe data workflows.

### Too Much Media LLC - Computer Programmer

Morganville, NJ | January 2016 - April 2017

- Developed real-time browser chat integrated with ticketing systems and implemented OAuth 2.0 authorization for protected sites.

### MFCXY, Inc. - Computer Programmer

Chicago, IL | December 2014 - September 2015

- Developed backend and Windows client features and middleware for cross-platform database migration.

### AJPM, LLC - Programmer / Owner

New Jersey / Remote | November 2012 - December 2013

- Built automated purchasing, inventory, repair, resale, and financial workflow software.

### ISPrime Inc. - CIO / Partner

Weehawken, NJ | March 2001 - December 2014

#### Infrastructure Leadership

- Architected and operated compute, hosting, and CDN infrastructure spanning approximately 3,000-4,000 servers, 10 locations, multiple datacenters, and more than 65 Gbps of peak traffic.

#### Systems Software Engineering

- Designed and built PrimeHTTPD, a high-performance, non-blocking, kqueue-based HTTP/CDN runtime in C for FreeBSD; deployed it across approximately 200 servers and supported more than 150,000 concurrent connections.
- Debugged performance and operational issues across the runtime and operating-system boundary, using sendfile(), SF\_NODISKIO, TCP\_NODELAY, TCP\_NOPUSH, O\_NONBLOCK, accept\_filter\_http, persistent connections, and dedicated I/O workers to keep disk operations from blocking the networking event loop.
- Developed production FreeBSD kernel modules including VirtualDir, KeepClean, and StatCache.

#### Security and Networking

- Built DDoS detection and mitigation systems, packet and DNS analysis tools, monitoring, telemetry, authentication, deployment, and security software used across production infrastructure.

### FatWallet.com - Systems Administrator

Wisconsin / Remote | December 2000 - June 2001

- Rebuilt production infrastructure to resolve performance and stability issues without customer-facing downtime and created automated availability and health alerting.

## EDUCATION

**Brookdale Community College** - Computer Science coursework; left during first semester to pursue ISPrime full-time during rapid company growth.